

WYNAU, Switzerland

WMO: 066430

Lat: 47.2550N

Lon: 7.7875E

Elev: 1365

StdP: 13.99

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 20.2	(d) 10.4	(e) 9.8	(f) 18.5	(g) 15.3	(h) 12.5	(i) 22.9	(j) 19.4	(k) 34.5	(l) 17.3	(m) 36.8	(n) 5.9	(o) 60	(p) 0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.7	67.8	83.1	66.8	79.8	65.5	69.7	82.6	68.1	79.8	66.7	76.9	4.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	99.7	73.1	64.4	95.1	71.5	63.1	90.8	70.5	34.5	82.5	33.2	79.9	32.0	77.2	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.1	(b) 13.6	(c) 11.5	(e) 10.4	(f) 91.5	(g) 5.1	(h) 3.1	(i) 6.7	(j) 93.8	(k) 3.7	(l) 95.6	(m) 0.8	(n) 97.3	(o) -2.9	(p) 99.6		
(a) 16.1	(b) 13.6	(c) 11.5	(e) 9.7	(f) 73.0	(g) 5.2	(h) 1.8	(i) 6.0	(j) 74.3	(k) 3.0	(l) 75.4	(m) 0.1	(n) 76.4	(o) -3.7	(p) 77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	33.1	34.9	42.0	48.9	56.8	63.8	66.8	66.0	58.4	50.8	40.6	34.6
	DBStd	13.59	6.80	7.13	6.60	6.36	5.95	6.43	5.52	5.52	5.63	6.30	6.29	6.44
	HDD50	2144	524	423	255	96	12	1	0	0	4	67	284	478
	HDD65	5804	988	843	712	482	260	99	45	55	207	439	731	943
	CDD50	2079	0	0	8	64	224	414	520	495	257	93	4	0
	CDD65	266	0	0	0	0	7	63	101	85	10	0	0	0
	CDH74	2449	0	0	0	12	123	618	903	705	87	1	0	0
	CDH80	699	0	0	0	1	18	180	274	219	7	0	0	0
Wind	WSAvg	3.7	4.4	4.5	4.5	4.0	3.8	3.4	3.2	2.7	3.0	3.1	3.5	4.3
Precipitation	PrecAvg	43.0	2.9	2.7	2.8	3.2	4.2	4.4	4.6	4.3	3.5	3.4	3.5	3.7
	PrecMax	54.4	7.4	6.6	9.8	7.0	7.2	9.1	8.0	8.4	6.2	6.4	8.3	7.8
	PrecMin	29.6	0.6	0.9	0.9	0.4	1.3	1.0	1.5	0.7	1.7	0.4	0.2	0.0
	PrecStd	6.4	1.7	1.5	1.9	1.8	1.7	1.9	1.5	1.8	1.4	1.6	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	56.6	66.5	75.6	82.4	90.0	91.4	91.4	80.9	70.3	59.9	51.4
		MCWB	47.4	46.8	51.9	57.6	64.0	69.4	69.1	69.3	65.9	60.2	52.9	48.1
	2%	DB	48.7	51.7	62.0	70.9	77.3	85.1	86.6	85.5	76.4	66.3	54.5	48.6
		MCWB	45.0	44.9	50.0	55.5	61.3	67.2	67.8	67.8	64.0	58.6	50.4	46.0
	5%	DB	45.5	48.1	57.8	66.7	73.5	81.0	82.7	81.6	72.4	62.7	51.6	46.3
		MCWB	42.6	43.0	48.2	53.4	59.7	66.0	66.8	67.0	62.3	56.9	48.4	44.1
	10%	DB	42.6	44.9	53.5	62.4	69.5	77.0	79.1	77.4	68.4	59.6	49.2	43.8
		MCWB	40.3	41.1	46.4	51.3	57.8	64.2	65.1	65.4	60.3	55.6	46.7	41.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	48.4	53.2	59.0	66.1	71.7	71.4	71.4	67.9	61.5	53.9	49.1
		MCDB	51.3	54.3	63.2	73.9	78.7	85.3	85.6	84.0	77.5	67.5	57.7	50.7
	2%	WB	45.4	45.7	51.1	56.3	63.2	68.8	69.3	69.5	65.3	59.7	51.2	46.5
		MCDB	48.1	50.9	59.4	68.5	74.0	81.5	82.3	81.9	73.7	64.6	54.2	48.4
	5%	WB	42.8	43.6	49.1	54.3	61.0	67.0	67.8	68.0	63.3	57.9	49.0	44.3
		MCDB	45.2	47.8	56.0	64.8	70.8	78.1	79.5	78.9	70.3	61.6	51.2	46.1
	10%	WB	40.5	41.3	47.1	52.3	58.9	65.2	66.4	66.4	61.5	55.9	47.0	42.0
		MCDB	42.4	44.5	53.0	60.5	67.1	74.9	76.3	75.3	67.0	59.2	48.9	43.7
Mean Daily Temperature Range	5% DB	MDBR	8.7	13.2	17.7	19.8	19.1	19.7	19.8	19.1	17.4	13.9	9.9	8.0
		MCDBR	12.9	19.8	26.7	29.3	27.5	27.4	26.9	26.9	23.6	18.4	14.1	10.9
		MCWBR	9.5	13.0	15.1	14.8	13.0	11.3	10.2	10.7	11.9	10.9	9.8	8.5
	5% WB	MCDBR	11.7	18.3	23.8	26.9	24.5	24.8	24.2	23.8	21.4	17.2	13.3	10.0
		MCWBR	9.2	12.4	14.0	13.8	12.0	10.6	9.8	10.4	11.1	10.6	9.7	8.3
Clear-Sky Solar Irradiance	taub	0.294	0.319	0.353	0.388	0.400	0.403	0.405	0.397	0.374	0.351	0.317	0.291	
	taud	2.446	2.380	2.339	2.275	2.287	2.318	2.308	2.344	2.388	2.440	2.478	2.457	
	Ebn at Noon	253	268	274	274	274	273	271	269	264	253	241	240	
	Edh at Noon	21	28	34	39	40	39	39	36	32	26	21	19	
All-Sky Solar Radiation	RadAvg	374	640	1024	1380	1589	1866	1800	1566	1193	755	413	304	
	RadStd	46	95	151	205	155	173	160	149	115	75	52	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.97	+0.94	N/A	N/A	+115	+65
(47) Regional (118 neighbors)	N/A	N/A	N/A	+1.33	+0.74	+0.56	N/A	N/A	+101	+27

Nomenclature: See separate page